



UNIVERSITY OF SALENTO
Department of Innovation Engineering

MASTER'S DEGREE IN AEROSPACE ENGINEERING

Individual project in:
Aerodynamics (Mod. 1)

DESIGN OF A SAILPLANE WING

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1 Introduction and methodology

This project aims at the aerodynamic sizing and the evaluation of the structural loads of the main wing of a sailplane, neglecting the presence of other surfaces such as the fuselage and control surfaces, using the methods acquired during the Aerodynamics (Mod. 1) course. The design specifications require the generation of a total lift equal to 300 kg ($L \approx 2942.5$ N) during a cruise phase at an altitude of 5000 m and at a flight speed $V_\infty = 40 \frac{\text{m}}{\text{s}}$, respecting a geometric constraint on the maximum allowable wing aspect ratio ($AR \leq 11$) [1].

The aerodynamic analyses were conducted under the assumption of potential flow (inviscid and irrotational fluid), assuming a steady flow regime; in any case neglecting viscosity and thus parasite drag. To justify the validity of the mathematical model, the fluid compressibility was evaluated: by extracting the thermodynamic properties at 5000 m of altitude using the International Standard Atmosphere (ISA) model [2, 3], the local speed of sound is equal to $a \approx 320.3 \frac{\text{m}}{\text{s}}$. The cruise Mach number is therefore:

$$M = \frac{V_\infty}{a} = \frac{40}{320.3} \approx 0.125 \quad (1.1)$$

Since the Mach number is well below the critical threshold ($M \ll 0.3$), compressibility effects are considered negligible and air density is assumed constant throughout the flow field. Furthermore, it is necessary to verify that the boundary layer is thin, to confirm that the irrotational flow assumption is plausible. For this reason, the Reynolds number was calculated [4]:

$$\text{Re} = \frac{\rho_\infty V_\infty \bar{c}}{\mu} = \frac{V_\infty \bar{c}}{\nu} \approx 2.215 \cdot 10^6 \quad (1.2)$$

Where $\bar{c} = \frac{1}{S} \int_{-b/2}^{b/2} [c(y)]^2 dy \approx 1.225$ m is the mean aerodynamic chord of the chosen designed finite wing (see Section 3.2), while the kinematic viscosity also obtained from the International Standard Atmosphere is: $\nu \approx 2.212 \cdot 10^{-5} \frac{\text{m}^2}{\text{s}}$. Given that the boundary layer thickness is $\delta \propto \frac{1}{\sqrt{\text{Re}}}$, the irrotational flow assumption is perfectly applicable over the entire flow field [4], moreover, the boundary layer was not evaluated here (consequently, the parasite drag will be zero and the overall aerodynamic efficiency of the wing cannot be calculated).

To tackle the design problem systematically, the workflow was divided into two sequential methodologies:

1. **Two-Dimensional (2D) Analysis:** The provided airfoil section was initially characterized using an aerodynamic solver based on the second-order *Vortex Panel Method* (VPM) [4, 5]. The use of vortex sheets with linearly varying intensity on the panels allowed for the high-fidelity extraction of pressure distributions and the airfoil's characteristic curves ($C_l - \alpha$ and $C_m - \alpha$). The obtained values were compared with those of the classical *Thin-Airfoil-Theory* (TAT) [4, 6] since the airfoil falls within its assumptions, but we know that the VPM is more accurate.
2. **Three-Dimensional (3D) Analysis:** The results of the 2D section were integrated into a code based on the *Non-Linear Lifting Line Theory* (NL-LLT) [4, 7]. Through the iterative resolution of the non-linear system, it was possible to evaluate different planform architectures, optimize the root AOA of the chosen configuration, and extract its overall aerodynamic performance and structural loads.

2 Aerodynamic characteristics of the airfoil

For the characterization of the provided airfoil [8], a code based on the second-order *Vortex Panel Method* [5] was developed, as described in [Section 1](#). The solver was validated using the analytical Joukowski airfoil (with an error on L' of less than 2%) and allows calculating the flow field and pressure distributions as the angle of attack varies.

2.1 Geometric processing

The original points of the airfoil, provided in counterclockwise order, were reordered and subjected to a cubic B-spline interpolation, to increase the number of points and thus the accuracy. A non-linear panel discretization was implemented. Cubic B-spline parameter:

$$t_{\text{non uniform}} = t - \frac{\sin(4\pi t)}{4\pi} \quad t \in [0; 1] \quad (2.1)$$

This approach allowed increasing the density of centroids near the leading edge and the trailing edge; running several simulations, it was observed that using the non-linear grid with over 500 points, the C_l remains the same in the first 5 significant digits. If a linear grid is used (utilized to derive the geometric characteristics with 2000 points), the number of points required to achieve the same behavior increases significantly, while the value converges to the one calculated using the non-linear grid. Once the interpolation was performed, the influence matrix and various geometric properties of the airfoil, listed and shown in [Figure 2.1](#), were calculated numerically. Everything was calculated normalizing with respect to the airfoil chord (the set of points was already provided assuming a unit chord).

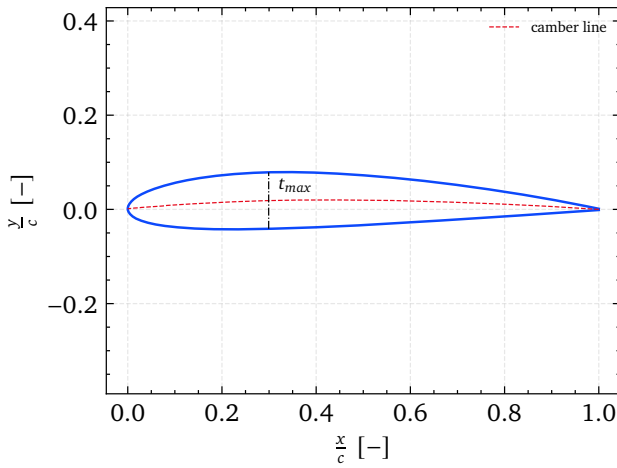


Figure 2.1: Interpolated geometry and numerically calculated geometric properties

2.2 Aerodynamic performance

By setting a freestream velocity of $40 \frac{\text{m}}{\text{s}}$ and solving the linear system to find the vector $\vec{v}$ for each angle of attack $\alpha \in \{-10, -9.75, -9.5, \dots, 9.5, 9.75, 10\}$ deg and calculating the tangential velocity at each centroid, the pressure coefficient C_p distribution was calculated using Bernoulli's theorem [4] at each angle of attack ([Figure 2.2](#)).

$$C_p = \frac{p - p_\infty}{\frac{1}{2} \rho V_\infty^2} = 1 - \left(\frac{V_{i,s}}{V_\infty} \right)^2 \quad (2.2)$$

A numerical limit was encountered: since the linear intensity distribution of the single vortex

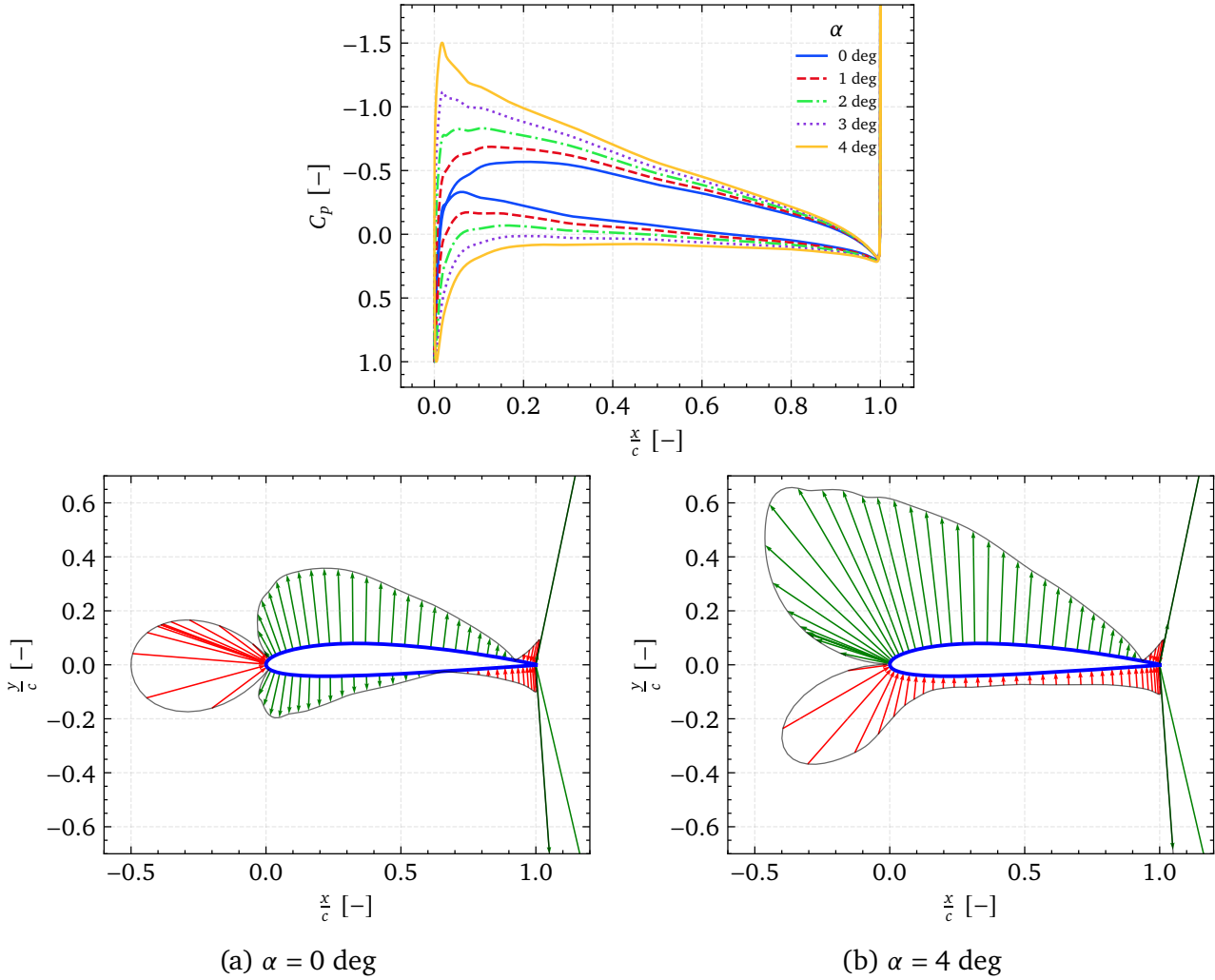


Figure 2.2: C_p distribution on the suction & pressure side of the airfoil as α varies

line handles sharp corners with difficulty, in a very small neighborhood of the trailing edge, the C_p diverges towards very high negative values. This phenomenon is less pronounced if a linear grid is used, but still tends to appear when significantly increasing the number of points. Nevertheless, the calculation is accurate because this divergence is localized in a very small neighborhood and occurs on both the suction and pressure sides. We proceed to calculate the C_l using the Kutta-Joukowski theorem [4], but first we need the total circulation (positive if clockwise), which we find by numerically integrating the intensity of the infinitesimal free vortices along the entire grid. The result is:

$$L' = \rho V_\infty \Gamma \quad \Rightarrow \quad C_l = \frac{L'}{0.5 \rho V_\infty^2 c} = \frac{\Gamma}{0.5 V_\infty c} \quad (2.3)$$

As for the C_m , assuming the simplifying hypothesis that the C_p is constant over the single vortex line (an assumption justified by the high density of the discretization), it will be:

$$dM_{LE \text{ nose-up}} = y_{arm} dF_x - x_{arm} dF_y \quad \Rightarrow \quad C_m = \frac{1}{c^2} \sum_j dC_m = \sum_j C_p \frac{x_{arm} dx + y_{arm} dy}{c^2} \quad (2.4)$$

Clearly, the value depends on the chosen reduction point; it was calculated for each α at the quarter-chord, while the position of the aerodynamic center [4] was also calculated numerically

by evaluating the derivatives of C_l and C_m , along with the value of $C_{m,x_{ac}}$ as shown in Table 2.1. The values of α , $C_l(\alpha)$, and $C_{m,c/4}(\alpha)$ were saved in a .dat file to be subsequently used in the NL-LLT as described in Section 3; let us observe the plots in Figure 2.3. From the linear regression of the numerical results, the characteristic aerodynamic parameters of the airfoil were extracted and compared with the theoretical value calculated using the *Thin-Airfoil-Theory* implementation [4, 6] in Table 2.1. The parasite drag per unit span will be zero due to *D'Alembert's Paradox* [4].

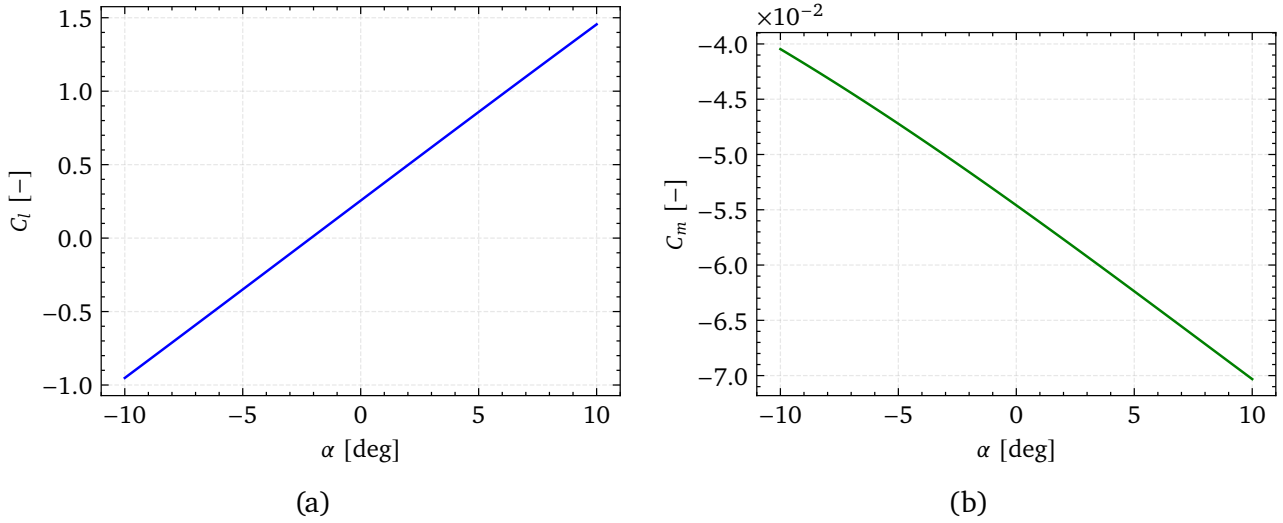


Figure 2.3: (a) C_l vs α , (b) $C_{m,c/4}$ vs α

Aerodynamic parameter	VPM	TAT
C_{l0}	0.2541	0.2317
$\frac{dC_l}{d\alpha}$	6.908	6.283
$\alpha_{L=0}$	-2.11°	-2.11°
$C_{m,c/4}$ (at $\alpha = 0$)	$-5.487 \cdot 10^{-2}$	$-5.298 \cdot 10^{-2}$
$\frac{dC_{m,c/4}}{d\alpha}$	$-8.620 \cdot 10^{-2}$	0
x_{ac}/c	0.2625	0.2500
$C_{m,x_{ac}}$	$-5.141 \cdot 10^{-2}$	TAT ($C_{m,c/4}$)

Table 2.1: Comparison of the airfoil's aerodynamic characteristics between VPM and TAT.

3 Design and performance of the finite wing

Having established the aerodynamic characteristics of the two-dimensional section, we proceeded with the design of the finite wing to meet the required flight condition described in [Section 1](#). The atmospheric properties ($\rho = 0.7361 \frac{\text{kg}}{\text{m}^3}$, $T = 255.65 \text{ K}$, $\mu = 1.628 \cdot 10^{-5} \frac{\text{kg}}{\text{m}\cdot\text{s}}$) were modeled by implementing the International Standard Atmosphere (ISA) [\[3\]](#).

3.1 Preliminary sizing and constraints

For the planform sizing, we operated within the limit imposed on the wing aspect ratio ($AR \leq 11$) and drew inspiration from sporting regulations for light sailplanes (FAI 13.5 Metre Class [\[9\]](#)). By setting the wingspan to $b = 13 \text{ m}$ and the aspect ratio to the maximum allowable value $AR = 11$, a reference wing area $S = 15.36 \text{ m}^2$ is obtained. This geometry requires an overall cruise lift coefficient equal to:

$$C_L = \frac{L}{\frac{1}{2} \rho V_\infty^2 S} \approx 0.325 \quad (3.1)$$

Which is perfectly consistent with experience; in fact, in cruise flight, we expect to fly at high speeds, thus trimmed at low C_L values. Furthermore, this leaves room for higher angles of attack during takeoff and landing, as it will require a very small α_r in cruise, as we will see in [Section 3.2](#).

3.2 Parametric analysis of the planform

To evaluate the wing's performance, a solver based on the *Non-Linear Lifting Line Theory* (NL-LLT) was developed [\[7\]](#). The non-linear system was solved iteratively to calculate the Fourier series coefficients of the circulation Γ . To ensure high accuracy, the simulation was set to 100 control points. In order to determine the optimal configuration, four architectures were analyzed ([Figures 3.1](#) and [3.2](#)):

1. **Wing 1 (Design configuration):** Tapered wing ($\lambda = 0.5$) with geometric twist of -1° .
2. **Wing 2 (Rectangular):** $\lambda = 1.0$, no twist.
3. **Wing 3 (Highly tapered):** $\lambda = 0.4$, no twist.
4. **Wing 4 (Elliptical):** Elliptical chord distribution (theoretical optimum).

The solver was coupled with an optimization algorithm to determine the root angle of attack α_r necessary to reach the lift target for each configuration. The numerical results and integrated loads are summarized in [Table 3.1](#).

As predicted by theory [\[4\]](#), the rectangular wing (Wing 2) has the lowest Oswald efficiency number ($e = 0.921$) and generates strong induced angle of attack (α_i) peaks at the wingtips. The elliptical wing (Wing 4) minimizes the induced angle of attack ($e = 1$) by uniformly distributing the load, but its structural manufacturing is complex. The design configuration (**Wing 1**) represents the ideal compromise: thanks to the 50 % taper ratio, an aerodynamic efficiency almost equal to the elliptical one and a reduced D_i are obtained. Furthermore, the adoption of the -1° washout unloads the wingtips, preventing tip stall. This effect translates into the lowest root bending moment among all the analyzed configurations (3976.4 Nm).

Configuration	α_r [deg]	e [-]	D_i [N]	$M_{f,\max}$ [Nm]	$M_{t,\max}$ [Nm]
Wing 1 (Tapered $\lambda = 0.5$)	1.59	0.9805	28.2	3976.4	-291.6
Wing 2 (Rectangular)	1.24	0.9211	30.1	4419.0	-287.1
Wing 3 (Tapered $\lambda = 0.4$)	1.15	0.9832	28.2	4058.0	-296.2
Wing 4 (Elliptical)	1.12	1.0000	27.7	4052.9	-305.5

Table 3.1: Comparison of aerodynamic performance and structural loads at the root for the four wing configurations.

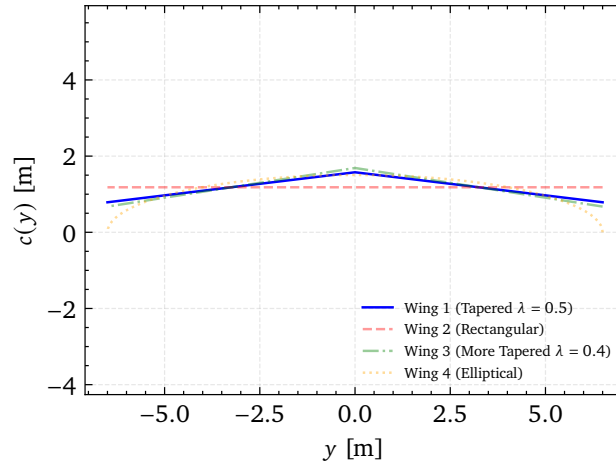


Figure 3.1: Spanwise planform chord distribution

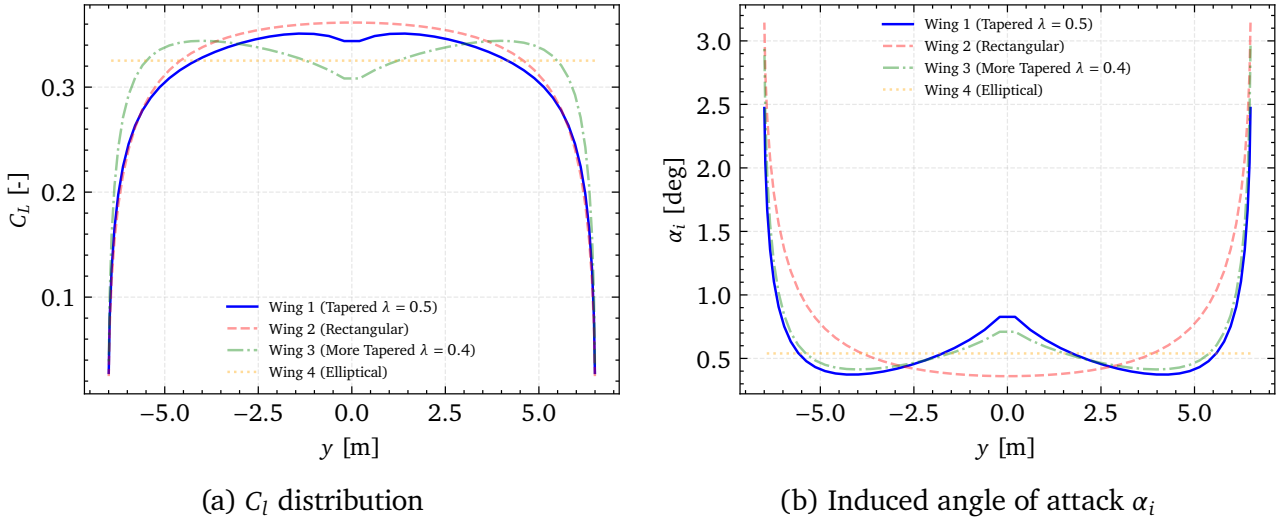


Figure 3.2: Aerodynamic comparison of the four analyzed configurations.

3.3 Global characteristic curves

For the final configuration (**Wing 1**), the root angle of attack α_r was varied between $\pm 6^\circ$. The global results (Figure 3.3) confirm a linear trend for the total lift L and the pitch moment M , and a parabolic trend for the induced drag D_i , as expected from experience, thus validating the robustness of the implemented mathematical model.

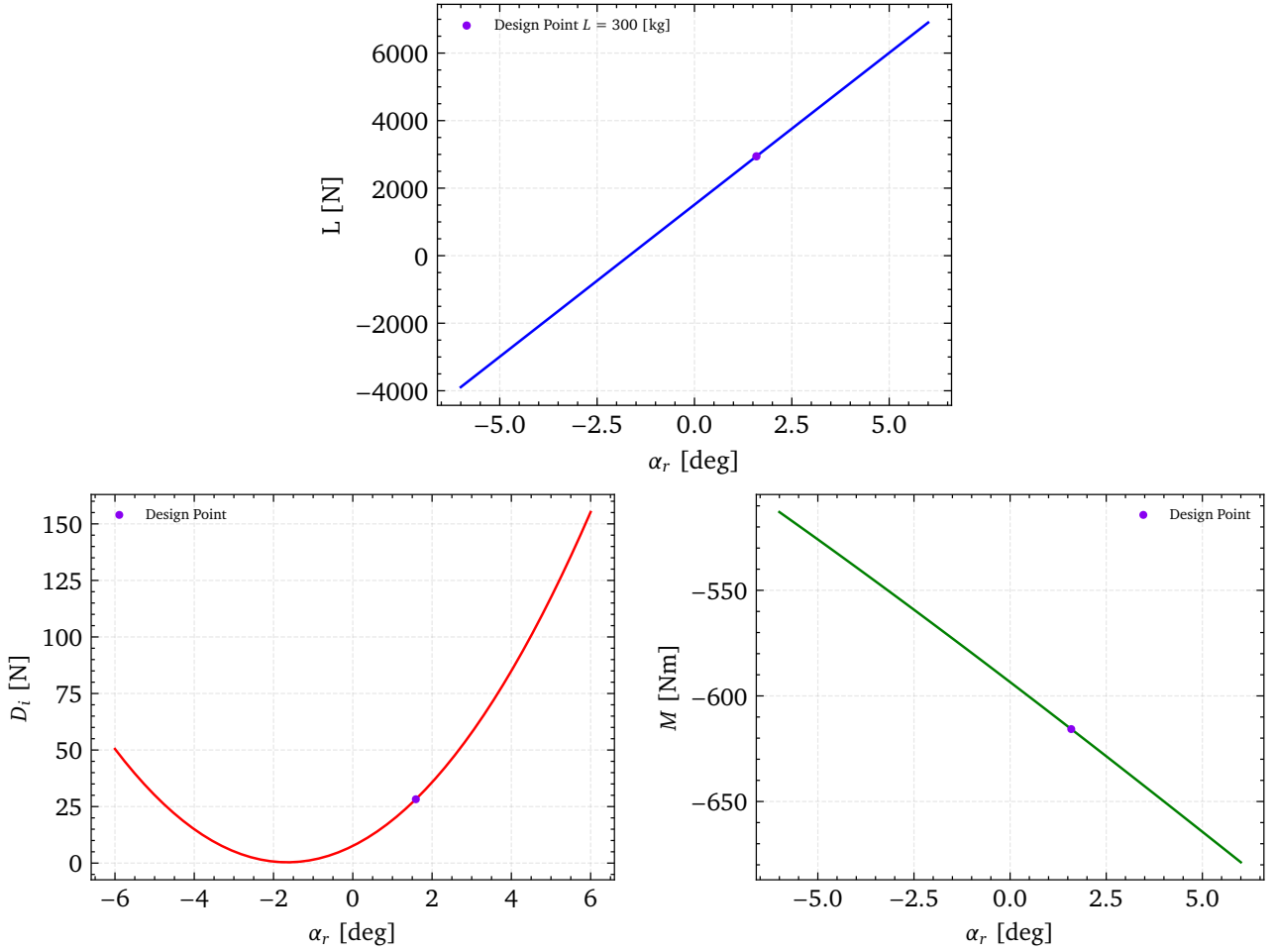


Figure 3.3: Global characteristic curves for the **Wing 1** configuration as a function of α_r .

If the parasite drag per unit span of the airfoil had also been calculated as the angle of attack varied, in this phase it would have been possible to integrate it in the spanwise direction, project it onto the freestream direction, and identify the total parasite drag.

3.4 Structural loads at the root

As requested by the assignment, the maximum loads at the wing root were calculated. For the design flight condition ($\alpha_r = 1.59^\circ$), the stresses were derived by numerically integrating [7] (via Simpson's rule) the aerodynamic load distributions along the semi-span ($y \in [0, b/2]$):

- **Maximum root bending moment ($M_{f,\max}$):** generated by the lift distribution $l(y) = \rho V_\infty \Gamma(y)$, it acts in the vertical plane and is evaluated as:

$$M_{f,\max} = \int_0^{b/2} \rho V_\infty \Gamma(y) y dy = 3976.4 \text{ Nm} \quad (3.2)$$

- **Maximum root twisting moment ($M_{t,\max}$):** generated by the intrinsic section moment coefficient $C_{m,c/4}$ along the span, it tends to twist the wing nose-down:

$$M_{t,\max} = \int_0^{b/2} \frac{1}{2} \rho V_\infty^2 c^2(y) C_{m,c/4}(y) dy = -291.6 \text{ Nm} \quad (3.3)$$

The pitch moment $\mathcal{M}$ due to the wing is obtained by integrating the twisting contribution over the entire wingspan:

$$\mathcal{M} = \int_{-b/2}^{b/2} \frac{1}{2} \rho V_{\infty}^2 c^2(y) C_{m,c/4}(y) dy = -615.7 \text{ Nm} \quad (3.4)$$

4 Conclusions

This project has successfully completed the aerodynamic sizing and the estimation of structural loads for the main wing of a sailplane, fully respecting the required operational and geometric specifications (total lift of approximately 2942.5 N at 5000 m altitude, flight speed of $40 \frac{\text{m}}{\text{s}}$ and wing aspect ratio $AR \leq 11$). The preliminary checks on the Mach number ($M \approx 0.125$) and the Reynolds number ($Re \approx 2.215 \cdot 10^6$) confirmed the validity of the irrotational and incompressible flow assumption adopted for the entire study.

From a methodological point of view, the use of the second-order *Vortex Panel Method* (VPM), combined with a non-linear B-spline interpolation of the given point cloud, allowed for the high-fidelity extraction of the airfoil's aerodynamic characteristics (C_l , C_m , and aerodynamic center), overcoming the accuracy limits of the classical *Thin-Airfoil-Theory* (TAT). The two-dimensional data reliably fed the three-dimensional solver based on the *Non-Linear Lifting Line Theory* (NL-LLT).

The parametric analysis conducted on four different planform architectures highlighted how the design configuration (**Wing 1**, characterized by a taper ratio $\lambda = 0.5$ and a geometric twist of -1°) represents the optimal compromise. The taper ratio guarantees an Oswald efficiency number ($e = 0.9805$) very close to the theoretical optimum of the elliptical wing, effectively minimizing the induced drag ($D_i = 28.2 \text{ N}$). By combining the adoption of the -1° aerodynamic twist, it was possible to unload the lift at the wingtips, minimizing the root bending moment ($M_{f,\max} = 3976.4 \text{ Nm}$), which is the lowest among all the analyzed configurations.

The global characteristic curves obtained for the finite wing confirm the expected theoretical trends, with a linear variation of the total lift and pitch moment, and a parabolic variation of the induced drag as the root angle of attack varies.

In conclusion, the developed aerodynamic model proved to be fully suitable for preliminary sizing. Future developments of this work may include the modeling of the viscous boundary layer, in order to quantify the parasite drag (neglected here due to D'Alembert's Paradox) and accurately determine the overall aerodynamic efficiency ($E = L/D$) of the aircraft in its cruise condition.

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